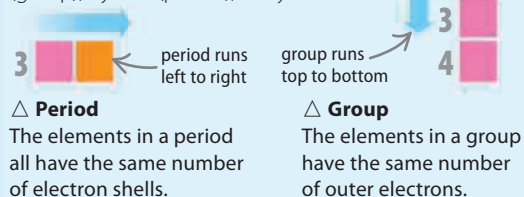


## Building blocks

The table can be divided by column (group), by row (period), or by series.



atoms of the elements in Group 18 have full outer shells

Only **63 elements** were known when **Dmitri Mendeleev** formulated the periodic table in 1869.

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→ 18

Key

- Alkali metals
- Alkali earth metals
- Transition metals
- Rare earth metals
- Other metals
- Metalloids
- Other nonmetals
- Halogens
- Noble gases
- Unknown

**Number 117**  
The latest element to be added, it has not been given a real name yet.

**New elements**  
These are often named after great scientists. Nobelium is named after Alfred Nobel.

|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
|-------------------------------------------|-------------------------------------------|---------------------------------------------|-------------------------------------------|---------------------------------------------|----------------------------------------------|----------------------------------------------|-----------------------------------------------|--------------------------------------------|---------------------------------------------|---------------------------------------------|--------------------------------------------|-----------------------------------------|--|--|--|--|--|----------------------------------------------|
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>2</b> 4.0026<br><b>He</b><br>HELIUM       |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>10</b> 20.180<br><b>Ne</b><br>NEON        |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>18</b> 39.948<br><b>Ar</b><br>ARGON       |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>36</b> 83.80<br><b>Kr</b><br>KRYPTON      |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>54</b> 131.29<br><b>Xe</b><br>XENON       |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>86</b> (222)<br><b>Rn</b><br>RADON        |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  | <b>118</b> (294)<br><b>Uuo</b><br>UNUNOCTIUM |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
| <b>9</b><br><b>Co</b><br>COBALT           | <b>10</b><br><b>Ni</b><br>NICKEL          | <b>11</b><br><b>Cu</b><br>COPPER            | <b>12</b><br><b>Zn</b><br>ZINC            | <b>13</b> 10.811<br><b>B</b><br>BORON       | <b>14</b> 12.011<br><b>C</b><br>CARBON       | <b>15</b> 14.007<br><b>N</b><br>NITROGEN     | <b>16</b> 15.999<br><b>O</b><br>OXYGEN        | <b>17</b> 18.998<br><b>F</b><br>FLUORINE   |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           | <b>31</b> 69.723<br><b>Ga</b><br>GALLIUM    | <b>32</b> 72.64<br><b>Ge</b><br>GERMANIUM    | <b>33</b> 74.922<br><b>As</b><br>ARSENIC     | <b>34</b> 78.96<br><b>Se</b><br>SELENIUM      | <b>35</b> 79.904<br><b>Br</b><br>BROMINE   |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           | <b>49</b> 114.82<br><b>In</b><br>INDIUM     | <b>50</b> 118.71<br><b>Sn</b><br>TIN         | <b>51</b> 121.76<br><b>Sb</b><br>ANTIMONY    | <b>52</b> 127.60<br><b>Te</b><br>TELLURIUM    | <b>53</b> 126.90<br><b>I</b><br>IODINE     |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           | <b>81</b> 204.38<br><b>Tl</b><br>THALLIUM   | <b>82</b> 207.2<br><b>Pb</b><br>LEAD         | <b>83</b> 208.96<br><b>Bi</b><br>BISMUTH     | <b>84</b> (209)<br><b>Po</b><br>POLONIUM      | <b>85</b> (210)<br><b>At</b><br>ASTATINE   |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           | <b>109</b> (268)<br><b>Mt</b><br>MEITNERIUM | <b>110</b> (281)<br><b>Ds</b><br>DARMSTADIUM | <b>111</b> (272)<br><b>Rg</b><br>ROENTGENIUM | <b>112</b> 15.999<br><b>Cn</b><br>COPERNICIUM | <b>113</b> 284<br><b>Uut</b><br>UNUNTRIUM  | <b>114</b> 289<br><b>Fl</b><br>FLEROVIUM    | <b>115</b> 288<br><b>Uup</b><br>UNUNPENTIUM | <b>116</b> 293<br><b>Lv</b><br>LIVERMORIUM | <b>117</b><br><b>Uus</b><br>UNUNSEPTIUM |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
| <b>62</b> 150.36<br><b>Sm</b><br>SAMARIUM | <b>63</b> 151.96<br><b>Eu</b><br>EUROPIUM | <b>64</b> 157.25<br><b>Gd</b><br>GADOLINIUM | <b>65</b> 158.93<br><b>Tb</b><br>TERBIUM  | <b>66</b> 162.50<br><b>Dy</b><br>DYSPROSIUM | <b>67</b> 164.93<br><b>Ho</b><br>HOLMIUM     | <b>68</b> 167.26<br><b>Er</b><br>ERBIUM      | <b>69</b> 168.93<br><b>Tm</b><br>THULIUM      | <b>70</b> 173.04<br><b>Yb</b><br>YTTERBIUM | <b>71</b> 174.97<br><b>Lu</b><br>LUTETIUM   |                                             |                                            |                                         |  |  |  |  |  |                                              |
|                                           |                                           |                                             |                                           |                                             |                                              |                                              |                                               |                                            |                                             |                                             |                                            |                                         |  |  |  |  |  |                                              |
| <b>94</b> (244)<br><b>Pu</b><br>PLUTONIUM | <b>95</b> (243)<br><b>Am</b><br>AMERICIUM | <b>96</b> (247)<br><b>Cm</b><br>CURIUM      | <b>97</b> (247)<br><b>Bk</b><br>BERKELIUM | <b>98</b> (251)<br><b>Cf</b><br>CALIFORNIUM | <b>99</b> (252)<br><b>Es</b><br>EINSTEINIUM  | <b>100</b> (257)<br><b>Fm</b><br>FERMIUM     | <b>101</b> (258)<br><b>Md</b><br>MENDELEVIUM  | <b>102</b> (259)<br><b>No</b><br>NOBELIUM  | <b>103</b> (262)<br><b>Lr</b><br>LAWRENCIUM |                                             |                                            |                                         |  |  |  |  |  |                                              |

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